Ergodicity of dynamical systems under stochastic noises

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Contents

Basic facts

- Metric dynamical system: $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ where $\theta : (\mathbb{R} \times \Omega, \mathcal{B}(\mathbb{R}) \otimes \mathcal{F}) \to (\Omega, \mathcal{F})$ such that $\theta_0 \equiv id_\Omega$ and $\theta_{t+s} = \theta_t \circ \theta_s$ and $\mathbb{P}$ is θ-invariant, i.e. $\mathbb{P}(\theta_t^{-1} A) = \mathbb{P}(A), \forall A \in \mathcal{F}, t \in \mathbb{R}$.
- θ is ergodic if all invariant sets, i.e. $A \in \mathcal{F}$ such that $\theta_t A = A \forall t \in \mathbb{R}$, have $\mathbb{P}(A) = 0$ or 1. This can be reduced to: every θ-invariant set mod $\mathbb{P}$ has measure 0 or 1, i.e. $\mathbb{P}(A \triangle \theta_t A) = 0, \forall t \in \mathbb{R}$.
- Lemma: θ is ergodic iff $\lim\limits_{t\to\infty} \frac{1}{t} \int_0^t \mathbb{P}(\theta_{-r} A \cap B) dr = \mathbb{P}(A)\mathbb{P}(B)$ for all $A, B \in \mathcal{A}$ such that $\sigma(\mathcal{A}) = \mathcal{F}$.
- Lemma: $T : (\Omega, \mathcal{F}) \to (\Omega, \mathcal{F})$ is measurable mapping, then define $T\mathbb{P}$-probability law of T w.r.t. $\mathbb{P}$ by $T\mathbb{P}(A) = \mathbb{P}(T^{-1}A)$. If for each $t \in \mathbb{R}$ there exists a set $\Omega_t \in \mathcal{F}$ of full measure such that $(\theta_t \circ T)_{|\Omega_t} = (T \circ \theta_t)_{|\Omega_t}$, then ergodicity of $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ leads to ergodicity of $(\Omega, \mathcal{F}, T\mathbb{P}, \theta)$.
- Examples: toss a coin $\Omega = (x_i)_{i\in\mathbb{Z}} : x_i \in \{H, T\}$, $\mathcal{F}$ generated from cylinder sets and $\mathcal{P}$ - Bernoulli measure, $\theta_t = T^i$ where $T(x_1, x_2, x_3, \ldots) = (x_2, x_3, \ldots)$ is the Bernoulli shift. Another example: Irrational rotation in the unit circle.
- Birkhorff's ergodic theorem: $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ is ergodic, $f \in L^1(\Omega, \mathbb{P})$, then for a.s. $\omega \in \Omega$

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t f(\theta_s \omega) ds = \int_\Omega f d\mathbb{P}.$$

Birkhorff-Khinchin theorem: if $T : \Omega \to \Omega$ is $\mathbb{P}$-measure preserving, and $f \in L^1(\Omega, \mathbb{P})$, $\mathcal{C}$ σ-algebra of all invariant sets of T then a.s. $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^n f(T^i \omega) = E(f|\mathcal{C})(\omega)$. (Varying limit!)

Stochastic stability: moment approach

- Asymptotic stability for deterministic systems: Lyapunov (1892) $\epsilon - \delta$ language.
- Stochastic stability for Markov systems: Khasminskii (1979), Skorohod (1987), ...
- Exponential stability: Molchanov, Arnold, Pardoux, Baxendale, Mao... and many others in control theory.

$$dy_t = [Ay_t + f(y_t)]dt + g(y_t)dB_t, \; x(0) = x_0 \in \mathbb{R}^d, \tag{1.1}$$

$A \in \mathbb{R}^{d\times d}$ neg. def. λ_A, $f : R^d \to \mathbb{R}^d$, $g : \mathbb{R}^d \to \mathbb{R}^{d\times m}$ glob. Lischitz continuous w.r.t. C_f, C_g.
Applying Itô's formula

$$d\|y_t\|^2 = \Big(2\langle y_t, Ay_t\rangle + 2\langle y_t, f(y_t)\rangle + \|g(y_t)\|^2\Big)dt + 2\langle y_t, g(y_t)dB_t\rangle \tag{1.2}$$

$$\leq (-2\lambda_A + 2C_f + C_g^2)\|y_t\|^2 + \|g(y_t)\|^2\Big)dt + 2\langle y_t, g(y_t)dB_t\rangle. \tag{1.3}$$

Applying the expectation, $\mathbb{E}\langle y_t, g(y_t)dB_t\rangle = 0$, thus

$$d\mathbb{E}\|y_t\|^2 \leq (-2\lambda_A + 2C_f + C_g^2)\mathbb{E}\|y_t\|^2 dt.$$

Applying continuous Gronwall lemma

$$\mathbb{E}\|y_t\|^2 \leq \mathbb{E}\|y_0\|^2 e^{-2(\lambda_A - C_f - \frac{1}{2}C_g^2)t}.$$

$\Rightarrow E\|y_t\|$ is exponentially decaying to zero given $\lambda_A - C_f > \frac{1}{2}C_g^2$, which follows that $\|y_t\|$ converges exponentially and almost surely to zero due to Borel-Catelli lemma.
Question: what happens if Brownian motion B is replaced by a fractional Brownian motion B^H?

Fractional Brownian motions

- Given $H \in (0,1)$, a continuous centered Gaussian process $B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s,t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t-s|^{2H}), \qquad s,\, t \in \mathbb{R},$$

  on an appropriate probability space, is called a two–sided one-dimensional fractional Brownian motion (fBm) with Hurst parameter H. When $H = 1/2$, B is the standard Brownian motion.

- Fractional Brownian motion B^H using the representation in Mandelbrot and van Ness

$$B^H(t) := \frac{1}{c_H}\Big(\int_{-\infty}^0 \big[(t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}}\big] dW(u) + \int_0^t (t-u)^{H-\frac{1}{2}} dW(u)\Big),$$

  with

$$c_H := \Big(\frac{1}{2H} + \int_0^\infty \big[(1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}}\big]^2 du\Big)^{\frac{1}{2}}.$$

- Kolmogorov-Centsov: for a.s. all realizations are Hölder continuous, i.e. $B^H_\cdot(\omega) = \omega_\cdot \in C^\beta([0,T],\mathbb{R}^m)$ for $\beta < H$. In fact, $\omega \in C^{0,\beta}([0,T],\mathbb{R}^m)$.
- FBM is not Markov, not even a semi-martingale, thus one can not apply Itô calculus to solve the equation.

Stochastic stability: pathwise approach

Pathwise approach

$$y_t = \Phi(t)y_0 + \int_0^t \Phi(t-s)f(y_s)ds + \int_0^t \Phi(t-s)g(y_s)dB_s$$

where $\Phi(t) = e^{At}$ is the semigroup generated by A. In addition, $\|\Phi(t)\| \leq C_A e^{-\lambda_A t}, \forall t \geq 0$. First, for any $t \in [n, n+1)$, it follows from (1.1) and the global Lipschitz continuity of f that

$$\begin{aligned}
\|y_t\| &\leq \|\Phi(t)y_0\| + \int_0^t \|\Phi(t-s)f(y_s)\|ds + \Big\|\int_0^t \Phi(t-s)g(y_s)dB_s\Big\| \\
&\leq C_A e^{-\lambda_A t}\|y_0\| + \int_0^t C_A e^{-\lambda_A(t-s)}C_f\|y_s\|ds + \Big\|\int_0^t \Phi(t-s)g(y_s)dB_s\Big\| \\
&\leq C_A e^{-\lambda_A t}\|y_0\| + C_A C_f \int_0^t e^{-\lambda_A(t-s)}\|y_s\|ds + \beta_t,
\end{aligned}$$

where $\beta_t := \Big\|\int_0^t \Phi(t-s)g(y_s)dB_s\Big\|$. Multiplying both sides of the above inequality with $e^{\lambda_A t}$ yields

$$\|y_t\|e^{\lambda_A t} \leq C_A\|y_0\| + \beta_t e^{\lambda_A t} + C_A C_f \int_0^t e^{\lambda_A s}\|y_s\|ds.$$

Stochastic stability: pathwise approach

By applying the continuous Gronwall Lemma, we obtain

$$\|y_t\|e^{\lambda_A t} \quad \leq \quad C_A\|y_0\| + \beta_t e^{\lambda_A t} + \int_0^t C_A C_f e^{C_A C_f (t-s)} \Big[C_A\|y_0\| + \beta_s e^{\lambda_A s}\Big] ds.$$

Once again, multiplying both sides of the above inequality with $e^{-C_A C_f t}$ and assigning $\lambda := \lambda_A - C_A C_f$ yields

$$\begin{aligned} \|y_t\|e^{\lambda t} \quad &\leq \quad C_A\|y_0\|e^{-C_A C_f t} + \beta_t e^{\lambda t} + \int_0^t C_A C_f e^{-C_A C_f s}\Big[C_A\|y_0\| + \beta_s e^{\lambda_A s}\Big] ds \\ &\leq \quad C_A\|y_0\| + \beta_t e^{\lambda t} + \int_0^t C_A C_f \beta_s e^{\lambda s} ds. \end{aligned} \tag{1.4}$$

Next, due to isometry of Itô integral

$$\mathbb{E}\beta_t^2 = \mathbb{E}\Big\| \int_0^t \Phi(t-s) g(y_s) dB_s \Big\|^2 = (C) \int_0^t \mathbb{E}\|\Phi(t-s)g(y_s)\|^2 ds \leq C_A^2 C_g^2 \int_0^t e^{-2\lambda_A (t-s)} \mathbb{E}\|y_s\|^2 ds.$$

$\Rightarrow$ isometry assumption can be relaxed upto a constant C, like G-Brownian motion!

Stochastic stability: pathwise approach

Then by applying the inequality

$$(a+b+c)^2 \leq (1+\epsilon)a^2 + (1+\frac{1}{\epsilon})(1+\bar{\epsilon})b^2 + (1+\frac{1}{\epsilon})(1+\frac{1}{\bar{\epsilon}})c^2, \quad \forall a,b,c,\epsilon,\bar{\epsilon} > 0$$

and

$$\Big(\int_0^t \|\beta_s\| e^{-\lambda(t-s)} ds\Big)^2 \leq \Big(\int_0^t e^{-2\varrho(t-s)} ds\Big)\Big(\int_0^t \|\beta_s\|^2 e^{-2(\lambda-\varrho)(t-s)} ds\Big)$$

to (1.4), we obtain for any $0 < \varrho < \lambda$ and any $\epsilon, \bar{\epsilon} > 0$ the estimate

$$\begin{aligned} \mathbb{E}\|y_t\|^2 &\leq (1+\frac{1}{\epsilon})(1+\frac{1}{\bar{\epsilon}})C_A^2 e^{-2(\lambda-\varrho)t}\mathbb{E}\|y_0\|^2 + (1+\epsilon)C_A^2C_g^2\int_0^t e^{-2(\lambda-\varrho)(t-s)}\mathbb{E}\|y_s\|^2 ds \\ &\quad +(1+\frac{1}{\epsilon})(1+\bar{\epsilon})C_A^2C_g^2\frac{L_f^2}{2\varrho}\int_0^t\int_0^s e^{-2(\lambda-\varrho)(t-u)}\mathbb{E}\|y_u\|^2 du\, ds. \qquad (1.5) \end{aligned}$$

Multiplying both sides of (1.5) by $e^{2(\lambda-\varrho)t}$ and assigning $\gamma_t := e^{2(\lambda-\varrho)t}\mathbb{E}\|y_t\|^2$, we derive an inequality

$$\gamma_t \leq (1+\frac{1}{\epsilon})(1+\frac{1}{\bar{\epsilon}})C_A^2\gamma_0 + (1+\epsilon)C_A^2C_g^2\int_0^t \gamma_s ds + (1+\frac{1}{\epsilon})(1+\bar{\epsilon})C_A^2C_g^2\frac{L_f^2}{2\varrho}\int_0^t\int_0^s \gamma_u du\, ds.$$

Stochastic stability: pathwise approach

Lemma (Gronwall lemma for iterated integrals)

Assume that $T, a, b, c > 0$ and $\gamma_t > 0$ satisfies

$$\gamma_t \leq a\gamma_0 + b\int_0^t \gamma_s ds + c\int_0^t\int_0^s \gamma_u du\, ds, \quad \forall t \in [0,T]. \tag{1.6}$$

Then $\gamma_t \leq a\gamma_0 e^{\nu_2 t}$ for all $t \in [0,T]$, where $\nu_2 > 0 > \nu_1$ are two roots of the quadratic $\nu^2 - b\nu - c = 0$.

It then follows from the iterated Gronwall lemma that $\gamma_t < (1+\frac{1}{\epsilon})(1+\frac{1}{\bar{\epsilon}})C_A^2\gamma_0 e^{\nu_2 t}$, where $\nu_2 = \nu_2(\varrho,\epsilon,\bar{\epsilon})$ is the positive root of the quadratic equation

$$\nu^2 - (1+\epsilon)C_A^2C_g^2\nu - (1+\frac{1}{\epsilon})(1+\bar{\epsilon})C_A^2C_g^2\frac{L_f^2}{2\varrho} = 0. \tag{1.7}$$

Hence $\mathbb{E}\|y_t\|^2 \leq (1+\frac{1}{\epsilon})(1+\frac{1}{\bar{\epsilon}})C_A^2\mathbb{E}\|y_0\|^2 e^{-2(\lambda-\varrho-\frac{\nu_2}{2})t}$, which derives the sufficient condition for the exponential stability as follows

$$\lambda = \lambda_A - C_AC_f > \inf_{\substack{0<\varrho<\lambda,\\ \epsilon>0}} \Big\{\varrho + \frac{1}{4}(1+\epsilon)C_A^2C_g^2 + \frac{1}{4}C_A^2C_g\Big[(1+\epsilon)^2C_g^2 + \frac{2C_f^2}{\varrho}(1+\frac{1}{\epsilon})\Big]^{\frac{1}{2}}\Big\}. \tag{1.8}$$

$\Rightarrow$ compare with $\lambda = \lambda_A - C_AC_f > \frac{1}{2}C_g^2$: still good enough with relaxed Itô isometry!

Stochastic stability: pathwise approach

How to estimate $\left\| \int_0^t \Phi(t-s)g(y_s)dB_s \right\|$ in general? Answer: rough path theory!

(i) The following estimate holds: for any $0 \leq a < b \leq c$

$$\left\| \int_a^b \Phi(c-s)g(y_s)dx_s \right\| \leq e^{-\lambda_A(c-b)}\kappa(\mathbf{x},[a,b])\Big(\|y_a\| + \left\|\!\left| y, R^y \right|\!\right\|_{p-\mathrm{var},[a,b]}\Big), \tag{1.9}$$

where

$$\kappa(\mathbf{x},[a,b]) := 2C_pC_A[1+\|A\|(b-a)]\Big\{C_g^2 \left\|\!\left|\mathbf{x}\right|\!\right\|^2_{p-\mathrm{var},[a,b]} \vee C_g \left\|\!\left|\mathbf{x}\right|\!\right\|_{p-\mathrm{var},[a,b]}\Big\}. \tag{1.10}$$

(ii) $z_t = \bar{y}_t - y_t$. Then for any $0 \leq a < b \leq c$

$$\left\| \int_a^b \Phi(c-s)[g(\bar{y}_s) - g(y_s)]dx_s \right\| \leq e^{-\lambda_A(c-b)}\kappa(\mathbf{x},[a,b])\Lambda(\mathbf{x},[a,b])\Big(\|z_a\| + \left\|\!\left| z, R^z \right|\!\right\|_{p-\mathrm{var},[a,b]}\Big), \tag{1.11}$$

for some function Λ. $\left\|\!\left| z, R^z \right|\!\right\|_{p-\mathrm{var},[a,b]}$ can be estimated by z_a.

(iii) $$\|y_t\|e^{\lambda t} \leq C_A\|y_0\| + e^{\lambda_A}\sum_{k=0}^n e^{\lambda k}\kappa(\mathbf{x},\Delta_k)\Big(\|y_k\| + \left\|\!\left| y, R^y \right|\!\right\|_{p-\mathrm{var},\Delta_k}\Big), \quad \forall t \in \Delta_n$$

where $\Delta_k := [k, k+1]$, $\lambda := \lambda_A - C_AC_f$, and $\left\|\!\left| y, R^y \right|\!\right\|_{p-\mathrm{var},\Delta_k}$ depends linearly on y_k.

Stochastic stability: pathwise approach

Assign $a := C_A\|y_0\|$, $u_k := \|y_k\|e^{\lambda k},\ k \geq 0$, one obtains for $G(\mathbf{x},\Delta_k) = G(C_g \;|\!|\!|\mathbf{x}|\!|\!|_{p-\mathrm{var},\Delta_k})$

$$u_n \leq a + \sum_{k=0}^{n-1} G(\mathbf{x},\Delta_k)u_k. \tag{1.12}$$

Discrete Gronwall Lemma: Let a be a non-negative constant and u_n, α_n, β_n be non-negative sequences satisfying $u_n \leq a + \sum_{k=0}^{n-1} \alpha_k u_k,\ \ \forall n \geq 1$. Then $u_n \leq \max\{a, u_0\}\prod_{k=0}^{n-1}(1+\alpha_k)$. Hence

$$\|y_n\| \leq C_A\|y_0\|e^{-\lambda n}\prod_{k=0}^{n-1}\Big[1+G(\mathbf{x},\Delta_k)\Big]$$

$$\Rightarrow \quad \lim_{n\to\infty}\frac{1}{n}\log\|y_n\| \leq -\lambda + \lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\log\Big[1+G(\mathbf{x},\Delta_k)\Big]. \tag{1.13}$$

Key construction: need $(\Omega,\mathcal{F},\mathbb{P},\theta)$ so that $G(\mathbf{x},\Delta_k) = G(\theta_k\omega,[0,1])$. Then apply Birkhorff's ergodic theorem gives

$$\lim_{n\to\infty}\frac{1}{n}\log\|y_n\| \leq -\lambda + E\log\Big[1+G(\omega,[0,1])\Big] \leq -\lambda + EG(\omega,[0,1]) < 0$$

for sufficiently small $C_g \Rightarrow$ exponential stability criterion.

Norms and spaces

For $p \geq 1$, $C^{p-\mathrm{var}}([a,b],H)$: the Banach space with finite norm

$$\|u\|_{p,[a,b]} = \|u_a\| + |\!|\!|u|\!|\!|_{p,[a,b]} = \|u_a\| + |\!|\!|u|\!|\!|_{p,[a,b]} = \Big(\sup_{\Pi(a,b)} \sum_{[s,t]\in\Pi} |u_{s,t}|^p\Big)^{1/p} < \infty.$$

For $\beta \in (0,1)$, $C^{\beta-\mathrm{Hol}}([a,b],H)$ the Banach space equipped with norm

$$\|u\|_{\infty,\beta,[a,b]} := \|u\|_{\infty,[a,b]} + |\!|\!|u|\!|\!|_{\beta,[a,b]} = \|u\|_{\infty,[a,b]} + \sup_{a\leq s<t\leq b} \frac{\|u(t)-u(s)\|}{(t-s)^\beta}.$$

In the simplest form for $\beta \in (\frac{1}{3},\frac{1}{2})$, a couple $\mathbf{x} = (x,\mathbb{X})$, with $x \in C^\beta([a,b],H)$ and $\mathbb{X} \in C_2^{2\beta}([a,b]^2, H\otimes H)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u}\otimes x_{u,t}, \qquad \forall a \leq s \leq u \leq t \leq b. \tag{2.1}$$

We write $\mathbb{X}_{s,t} = (x\otimes x)_{s,t} =: \int_s^t x_{s,u}\otimes dx_u$. Define the semi-norm

$$|\!|\!|\mathbf{x}|\!|\!|_{p-\mathrm{var},[a,b]} = \Big(|\!|\!|x|\!|\!|_{p-\mathrm{var},[a,b]} + |\!|\!|\mathbb{X}|\!|\!|_{\frac{p}{2}-\mathrm{var},[a,b]}\Big)^{\frac{1}{p}}.$$

Examples in $1D$

- $X_t = \int_0^t a_s dB_s \Rightarrow \mathbb{X}_{s,t} := \int_s^t X_{s,u} dX_u = \frac{1}{2}X_{s,t}^2 - \frac{1}{2}\int_s^t a_u^2 du$. Integral in the Itô sense.
- $X_t = B^H \Rightarrow \mathbb{X}_{s,t} := \int_s^t B^H_{s,u}\delta B^H_u = \frac{1}{2}(B^H_{s,t})^2 - \frac{1}{2}\Big(t^{2H} - s^{2H}\Big)$. Integral in the Skorohod-Wick-Itô sense.

$\beta > \frac{1}{3}$: rough integrals

- Consider $y \in C^\beta([a,b], V\otimes H)$ and $x \in C^\beta([a,b],H)$, there exists a Young integral $\int_a^b ydx$ defined by
$$\int_a^b ydx = \lim_{|\Pi|\to 0} \sum_{[s,t]\in\Pi} y_s x_{s,t},$$
Young-Loeve (Y-L) estimate
$$\Big\| \int_s^t y_u dx_u - y_s x_{s,t}\Big\| \leq K_\beta |t-s|^{2\beta} \|y\|_{\beta,[a,b]} \|x\|_{\beta,[a,b]}.$$
- A path $y \in C^\beta([a,b], V\otimes H)$ is then called to be *controlled by* $x \in C^\beta([a,b],H)$ if there exists a tube (y', R^y) with $y' \in C^\beta([a,b], V\otimes H\otimes H)$, $R^y \in C^{2\beta}(\Delta^2([a,b]), V\otimes H)$ such that
$$y_{s,t} = y'_s x_{s,t} + R^y_{s,t}, \qquad \forall \min I \leq s \leq t \leq \max I.$$
Thanks to the sewing lemma, the rough integral $\int_s^t y_u dx_u$ can be defined as
$$\int_s^t y_u dx_u = \int_s^t y_u d\mathbf{x}_u := \lim_{|\Pi|\to 0} \sum_{[u,v]\in\Pi} [y_u x_{u,v} + y'_u \mathbb{X}_{u,v}]$$
Moreover, one gets Y-L typed estimate:
$$\Big\| \int_s^t y_u dx_u - y_s x_{s,t} - y'_s \mathbb{X}_{s,t}\Big\| \leq C_\beta |t-s|^{3\beta}\Big(\|x\|_{\beta,[s,t]} \big\|R^y\big\|_{2\beta,\Delta^2[s,t]} + \big\|y'\big\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]}\Big).$$

Existence and uniqueness theorem: RDEs

- Finite dimension: RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ is understood in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)d\mathbf{x}_s. \tag{2.2}$$

- Riedel and Scheutzow (2017): f is one sided Lipschitz and linear growth in perpendicular direction. $g \in C_b^3$ as usual. Solution of the form (y, y') with $y, y' \in C^\beta$, $y' = g(y)$, which is solved by Doss-Sussmann technique.
- Infinite dimension: consider rough evolution equation

$$dy_t = [Ay_t + f(y_t)]dt + g(y_t)dx_t, \tag{2.3}$$

with the solution understood in the mild form

$$y_t = S(t)y_0 + \int_0^t S(t-s)f(y_s)ds + \int_0^t S(t-s)g(y_s)d\mathbf{x}_s. \tag{2.4}$$

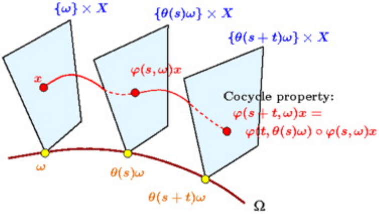
- Garrido-Atienza & Lu & Schmalfuss (2015): define rough integral using fractional calculus and solution definition by Hu & Nualart (2009).
- Hesse & Neamtu (2019): define rough integrals for controlled rough path $y \in C^{\beta,\beta}$ and $x \in C^\beta$ using a modified sewing lemma.

Random dynamical systems

Definition

A (continuous) random dynamical system is an NDS (θ, φ) which in addition has the following properties:
(i), The driving system θ is an ergodic dynamical system $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t\in T})$, i.e., the base $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $(t, \omega) \mapsto \theta_t\omega$ is a measurable flow which is ergodic under $\mathbb{P}$.
ii, The cocycle $(t, \omega, x) \mapsto \varphi(t, \omega)x$ is measurable w.r.t $\mathcal{B} \otimes \mathcal{F} \otimes \mathcal{B}(X)$ and $\mathcal{B}(X)$, respectively, where $\mathcal{B}(X)$ is the Borel $\sigma-$algebra of X, such that the family $\varphi(t, \omega, \cdot) = \varphi(t, \omega) : X \to X$ of self-mappings of X satisfies the *cocycle property*

$$\varphi(0, \omega) = id_X,\ \varphi(t + s, \omega) = \varphi(t, \theta_s\omega) \circ \varphi(s, \omega),\ \forall t, s \in T,\ \forall \omega \in \Omega. \qquad (3.1)$$



Generation of RDS: rough cocycles

- Finite dimension - Bailleul & Riedel & Scheutzow (2017)
- Chen relation leads to a certain relation like $(1, x_{s,t}, \mathbb{X}_{s,t}) = (1, x_{s,u}, \mathbb{X}_{s,u}) \bullet (1, x_{u,t}, \mathbb{X}_{u,t})$. How ?

$T_1^2(\mathbb{R}^m) = 1 \oplus \mathbb{R}^m \oplus (\mathbb{R}^m \otimes \mathbb{R}^m)$, is the set with the tensor product

$$(1, g^1, g^2)\bullet(1, h^1, h^2) = (1, g^1+h^1, g^1\otimes h^1+g^2+h^2), \quad \forall\, \mathbf{g} = (1, g^1, g^2), \mathbf{h} = (1, h^1, h^2) \in T_1^2(\mathbb{R}^m).$$

Then it can be shown that $(T_1^2(\mathbb{R}^m), \bullet)$ is a topological group with unit element $\mathbf{1} = (1, 0, 0)$. For $\beta \in (\frac{1}{p}, \nu)$, denote by $\mathcal{C}^{0,\beta}([a, b], T_1^2(\mathbb{R}^m))$ the closure of $\mathcal{C}^{\infty}([a, b], T_1^2(\mathbb{R}^m))$ in $\mathcal{C}^{\beta}([a, b], T_1^2(\mathbb{R}^m))$, and by $\mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ the space of all $x : \mathbb{R} \to \mathbb{R}^m$ such that $x|_I \in \mathcal{C}^{0,\beta}(I, T_1^2(\mathbb{R}^m))$ for each compact interval $I \subset \mathbb{R}$ containing 0. Then $\mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ is equipped with the compact open topology given by the $p-$variation norm, i.e the topology generated by the metric:

$$d_\beta(\mathbf{x}_1, \mathbf{x}_2) := \sum_{k\geq 1} \frac{1}{2^k}(\|\mathbf{x}_1 - \mathbf{x}_2\|_{\beta,[-k,k]} \wedge 1).$$

Let us consider a stochastic process $\bar{\mathbf{X}}$ defined on a probability space $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{\mathbb{P}})$ with realizations in $(\mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m)), \mathcal{F})$. Assume further that $\bar{\mathbf{X}}$ has stationary increments.

Generation of RDS: rough cocyles

Assign

$$\Omega := \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$$

and equip with the Borel $\sigma-$ algebra $\mathcal{F}$ and let $\mathbb{P}$ be the law of $\bar{\mathbf{X}}$. Denote by θ the *Wiener-type shift*

$$(\theta_t\omega)_\cdot = \omega_t^{-1} \bullet \omega_{t+\cdot}, \forall t \in \mathbb{R}, \omega \in \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m)), \tag{3.2}$$

and define the so-called *diagonal process* $\mathbf{X} : \mathbb{R} \times \Omega \to T_1^2(\mathbb{R}^m), \mathbf{X}_t(\omega) = \omega_t$ for all $t \in \mathbb{R}, \omega \in \Omega$. Due to the stationarity of $\bar{\mathbf{X}}$, it can be proved that θ is invariant under $\mathbb{P}$, then forming a continuous (and thus measurable) dynamical system on $(\Omega, \mathcal{F}, \mathbb{P})$ [BRSch17 -Theorem 5]. Moreover, $\mathbf{X}$ forms a *β-Hölder rough path cocycle*, namely, $\mathbf{X}_\cdot(\omega) \in \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ for every $\omega \in \Omega$, which satisfies the *cocyle relation*:

$$\mathbf{X}_{t+s}(\omega) = \mathbf{X}_s(\omega) \bullet \mathbf{X}_t(\theta_s\omega), \forall \omega \in \Omega, t, s \in \mathbb{R},$$

in the sense that $\mathbf{X}_{s,s+t} = \mathbf{X}_t(\theta_s\omega)$ with the increment notation $\mathbf{X}_{s,s+t} := \mathbf{X}_s^{-1} \otimes \mathbf{X}_{s+t}$. It is important to note that the two-parameter flow property

$$\mathbf{X}_{s,u} \bullet \mathbf{X}_{u,t} = \mathbf{X}_{s,t}, \forall s, t \in \mathbb{R}$$

is equivalent to the fact that $\mathbf{X}_t(\omega) = (1, x_t(\omega), \mathbb{X}_{0,t}(\omega))$, where $x_\cdot(\omega) : \mathbb{R} \to \mathbb{R}^m$ and $\mathbb{X}_{\cdot,\cdot}(\omega) : I \times I \to \mathbb{R}^m \otimes \mathbb{R}^m$ are random funtions satisfying Chen's relation relation (2.1).
How can we prove ergodicity of θ?

Ergodicity of θ: $X = B^H$

- $H = \frac{1}{2}$: $\Omega = C_0^0(\mathbb{R},\mathbb{R})$, $\mathcal{F}$ is σ-algebra is generated by the compact open topology. Then construct $B_t(\omega) = \omega_t$ the caconical process w.r.t. Wiener measure $\mathbb{P}_{\frac{1}{2}}$ and Wiener shift $\theta_t\omega_\cdot = \omega_{t+\cdot} - \omega_\cdot$. Then θ is ergodic.
- Garrido-Atienza, B. Schmalfuss (2011): the Wiener shift $\eta_t x_\cdot = x_{t+\cdot} - x_t$ is ergodic on $\Omega' = C^{0,\beta}(\mathbb{R},\mathbb{R}^m)$ w.r.t. $\mathbb{P}_H = B^H\mathbb{P}_{\frac{1}{2}}$.
- Friz-Hairer (2014, Theorem 10.4): there exists a full measure subset $\Omega_1 \subset \Omega'$ such that $\omega = (1, x, \mathbb{X}) \in \Omega$ for any $x \in \Omega_1$. Moreover, by Friz-Victoir (2010, Theorem 15.42, 45), one can choose this full measure subset Ω_1 such that it satisfies the piecewise linear approximations (mollifier approximation).
- Then consider the natural lift $\mathcal{S}$ on smooth paths

$$\mathcal{S}(x)_{s,t} = (1, x_{s,t}, \int_s^t x_{s,r} dx_r), \tag{3.3}$$

which can be extended to Ω_1 such that $\mathcal{S} : \Omega_1 \to \Omega$. By Bailleul, Riedel, Scheutzow (2017), there exists a metric dynamical system $(\hat\Omega, \hat{\mathcal{F}}, \hat{\mathbb{P}}, \theta)$ such that $\hat\Omega \subset \Omega$ and $\hat{\mathbb{P}} = \mathbb{P}^H \circ \mathcal{S}^{-1}$. Furthermore,

$$\theta_t\mathcal{S}(x) = \lim_{n\to\infty} \hat\theta\mathcal{S}(x^{(n)}) = \lim_{n\to\infty} \mathcal{S}(\eta_t x^{(n)}) = \mathcal{S}(\eta_t x). \tag{3.4}$$

- Since η is ergodic, by Garrido-Atienza, B. Schmalfuss (2011, Lemma 3), θ is also ergodic.

Generation of RDS from rough equations

In particular, due to the fact that $\|\mathbf{X}_\cdot(\theta_h\omega)\|_{p-\mathrm{var},[s,t]} = \|\mathbf{X}_\cdot(\omega)\|_{p-\mathrm{var},[s+h,t+h]}$, it follows from Birkhorff ergodic theorem that

$$\Gamma(\mathbf{x},p) := \limsup_{n\to\infty}\Big(\frac{1}{n}\sum_{k=1}^{n}\|\theta_{-k}\mathbf{x}\|^p_{p-\mathrm{var},[-1,1]}\Big)^{\frac{1}{p}} = \Big(E\,\|\mathbf{X}_\cdot(\cdot)\|^p_{p-\mathrm{var},[-1,1]}\Big)^{\frac{1}{p}} = \Gamma(p) \tag{3.5}$$

for almost all realizations $\mathbf{x}_t$ of the form $\mathbf{X}_t(\omega)$.

Theorem

Given the measurable metric dynamical system $(\Omega,\mathcal{F},\mathbb{P},\theta)$ *and the p- rough cocycle* $\mathbf{X}:\mathbb{R}\times\Omega\to T^2_1(\mathbb{R}^m)$ *as above, the system* (3.6) *generates a continuous random dynamical system* φ *over* $(\Omega,\mathcal{F},\mathbb{P},\theta)$*, such that for any* $[0,T]$ *and all* $\omega\in\Omega$*,* $\varphi(t,\omega)y_0$ *is the unique solution (in the Gubinelli sense) of* (2.4)*,*

$$dy_t = f(y_t)dt + g(y_t)d\mathbf{X}_t(\omega), t\geq 0, \tag{3.6}$$

on [0,T], where $\mathbf{x}=(x,\mathbb{X})$ *is the projection of* $\mathbf{X}_\cdot(\omega)$ *on* $\mathbb{R}^m\oplus(\mathbb{R}^m\otimes\mathbb{R}^m)$*.*

Random attractors

Universe: $\mathcal{D}$ a family of random sets $\hat{M} = \{M(\omega)\}_{\omega\in\Omega}$ which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$).

Definition (Crauel & Flandoli-1994)

An invariant random compact set $\hat{A} \in \mathcal{D}$ is called
(i), a pullback random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t\to\infty} d(\varphi(t,\theta_{-t}\omega)D(\theta_{-t}\omega)|A(\omega)) = 0; \tag{4.1}$$

(ii), a forward random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the forward sense, i.e.

$$\lim_{t\to\infty} d(\varphi(t,\omega)D(\omega)|A(\theta_t\omega)) = 0; \tag{4.2}$$

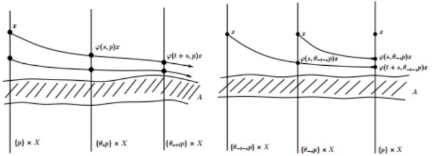


Figure 2.1: Forward attraction versus pullback attraction.

Existence of random attractor

Pullback absorbing set: $\hat{B} \in \mathcal{D}$ in a universe $\mathcal{D}$ such that $\hat{B}$ absorbs all sets in $\mathcal{D}$, i.e. for any $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega) \subset B(\omega), \text{ for all } t \geq t_0.$$

Theorem (Crauel, Flandoli, Schenk-Hoppe 1998: Existence and Uniqueness of Random Attractor)

Given a universe $\mathcal{D}$, assume there exists a random compact pullback absorbing set $\hat{B} \in \mathcal{D}$ which is forward invariant. Then there exists a unique random pullback attractor (which is then a weak attractor) in $\mathcal{D}$, given by

$$A(\omega) = \bigcap_{t\geq 0} \varphi(t, \theta_{-t}\omega)B(\theta_{-t}\omega).$$

Application: prove that a system has an absorbing set.

Random attractors for RDE: finite dimension

We would like to investigate the RDE of the form

$$dy_t = f(y_t)dt + g(y_t)dx_t. \tag{4.3}$$

($\mathbf{H}'_f$) and the dissipativity condition

$$\exists D_1 \geq 0, D_2 > 0 : \quad \langle y, f(y)\rangle \leq \|y\|(D_1 - D_2\|y\|), \quad \forall y \in \mathbb{R}^d, \tag{4.4}$$

Condition (4.4) is equivalent to the following condition: there exist constants $d_1 \geq 0, d_2 > 0$ such that

$$\langle y, f(y)\rangle \leq d_1 - d_2\|y\|^2, \quad \forall y \in \mathbb{R}^d; \tag{4.5}$$

($\mathbf{H}_g$) g belongs to $C^3_b(\mathbb{R}^d, (\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max\left\{\|g\|_\infty, \|Dg\|_\infty, \|D^2_g\|_\infty, \|D^3_g\|_\infty\right\} < \infty; \tag{4.6}$$

($\mathbf{H}_X$) for a given $\nu \in (\frac{1}{3}, \frac{1}{2})$, x belongs to the space $C^\nu(\mathbb{R}, \mathbb{R}^m)$ of all continuous paths which is of finite $\nu-$Hölder norm on any interval $[s, t]$. In particular, x is a realization of a stationary stochastic process $X_t(\omega)$, such that x can be lifted into a realized component $\mathbf{x} = (x, \mathbb{X})$ of a stationary stochastic process $(x_\cdot(\omega), \mathbb{X}_{\cdot,\cdot}(\omega))$, such that the estimate

$$E\Big(\|x_{s,t}\|^p + \|\mathbb{X}_{s,t}\|^q\Big) \leq C_{T,\nu}|t-s|^{p\nu}, \forall s, t \in [0, T] \tag{4.7}$$

holds for any $[0, T]$, with $p\nu \geq 1, q = \frac{p}{2}$ and some constant $C_{T,\nu}$.

Random attractors for RDE: finite dimension

($\mathbf{H}_{\mathcal{A}}$) There exists a duration $r > 0$ and constants $D_3 > 0$ of the deterministic system such that, for any starting point $y_0 \notin \mathcal{A}$, there exists a point $\mu_0 = \mu_0(y_0) \in \mathcal{A}$ satisfying

$$\|\mu_r(y_0) - \mu_r(\mu_0)\| \leq e^{-D_3}\|y_0 - \mu_0\|. \tag{4.8}$$

Theorem (LHD-DCDS 2022)

Assume that system (3.6) *satisfies the assumptions* $(\boldsymbol{H}_f'), (\boldsymbol{H}_g), (\boldsymbol{H}_X)$*, then there exists a random pullback attractor* $\mathcal{A}(\omega)$ *such that* $|\mathcal{A}(\cdot)| \in \mathcal{L}^\rho$ *for any* $\rho \geq 1$*. If in addition* $(\boldsymbol{H}_{\mathcal{A}})$ *and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that* $\mathcal{A}(\omega) \to \mathcal{A}$ *(w.r.t. the Hausdorff semi-distance) as* $C_g \to 0$*, both in the almost sure and in* $\mathcal{L}^\rho$ *senses. Moreover, if f is strictly dissipative then* $\mathcal{A}(\omega)$ *is a singleton provided that* C_g *is sufficiently small.*

Sketch of the proof: Doss-Sussmann technique to conjugate the RDE to a random differential equations. Under the assumptions ($\mathbf{H}_f'$), ($\mathbf{H}_g$), ($\mathbf{H}_{\mathcal{X}}$), ($\mathbf{H}_{\mathcal{A}}$) and $f \in Lip$, the random attractor is upper semi-continuous, i.e.

$$\lim_{C_g \to 0} d_H\Big(\mathcal{A}(\omega)|\mathcal{A}\Big)^\rho = 0 \quad \text{a.s.} \quad \text{and} \quad \lim_{C_g \to 0} \mathbb{E}\, d_H\Big(\mathcal{A}(\cdot)|\mathcal{A}\Big)^\rho = 0, \quad \forall \rho \geq 1. \tag{4.9}$$

RDS from Euler scheme

we consider the explicit Euler scheme for the regular grid with step size $h > 0$, i.e. $\Pi = \{kh\}_{k\in\mathcal{N}}$ and

$$\begin{aligned} &y_0^h \in \mathbb{R}^d, \\ &y_{k+1}^h = y_k^h + f(y_k^h)h + g(y_k^h)x_{kh,(k+1)h}(\omega) + Dg(y_k^h)g(y_k^h)\mathbb{X}_{kh,(k+1)h}(\omega), \quad k \in \mathcal{N}. \end{aligned} \tag{4.10}$$

Such a scheme is well defined. We rewrite (4.10) as

$$\begin{aligned} y_{k+1}^h &= \underbrace{\Big(y_k^h + f(y_k^h)h\Big)}_{=:F(h,y_k^h)} + \Big\langle \underbrace{\Big(g(y_k^h), Dg(y_k^h)g(y_k^h)\Big)}_{=:G(y_k^h)}, \Big(x_{kh,(k+1)h}(\omega), \mathbb{X}_{kh,(k+1)h}(\omega)\Big)\Big\rangle \\ &= F(h,y_k^h) + G(y_k^h)\bar{\otimes}\Big(1, x_{kh,(k+1)h}(\omega), \mathbb{X}_{kh,(k+1)h}(\omega)\Big) \\ &= F(h,y_k^h) + G(y_k^h)\bar{\otimes}\big(\theta_{kh}\omega\big)_h = H(h,\theta_{kh}\omega)y_k^h, \end{aligned} \tag{4.11}$$

where we introduce the generator function

$$H(h,\omega)y := F(y) + G(y)\bar{\otimes}\omega_h. \tag{4.12}$$

Discrete-time random dynamical system $\varphi^h : \mathcal{N}^h \times \Omega \times \mathbb{R}^d \to \mathbb{R}^d$ over $\mathcal{N}^h := \{kh\}_{k\in\mathcal{N}}$ and $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that for any $\omega \in \Omega$ and $y_0^h \in \mathbb{R}^d$, $\varphi^h(k,\omega)y_0^h$ is defined from (4.11) by

$$\begin{aligned} &\varphi^h(0,\omega)y_0^h \equiv y_0^h, \\ &\varphi^h(kh,\omega)y_0^h := y_k^h = H(h,\theta_{(k-1)h}\omega)\circ\ldots\circ H(h,\omega)y_0^h, \quad \forall k \geq 1. \end{aligned} \tag{4.13}$$

Random attractor for Euler scheme

Theorem (LHD, Kloeden, Cong, Hong (2021-2022))

Under the hypotheses ($\mathbf{H}_f'$),($\mathbf{H}_g^b$) and ($\mathbf{H}_X$), assume further that f is dissipative with (4.4) and of linear growth, i.e. there exists C_f such that

$$\|f(y)\| \leq C_f(1 + \|y\|). \tag{4.14}$$

Then there exists a $h_0 > 0$ such that for all $h < h_0$, the generated discrete-time random dynamical system φ^h from (4.11) admits a global random pullback attractor $A^h(\omega)$.

Theorem (LHD, Kloeden, Cong, Hong (2021-2022))

Assume ($\mathbf{H}_g^b$), ($\mathbf{H}_X$) with the Gaussian process X and further that f is globally Lipschitz continuous and bounded, such that the dissipativity condition (4.4) is satisfied. Then φ^h admits a numerical pullback attractor $\mathcal{A}^h$ which converges to the attractor $\mathcal{A}$ a.s. in the Hausdorf semi-distance, i.e.,

$$\lim_{h\to 0} d_H(\mathcal{A}^h|\mathcal{A}) = 0 \quad a.s. \tag{4.15}$$

Thank you!

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